Conceptual Design of 2016 ITS Starship

VTVL TSTO Reusable Launch System [1,2,3]

Faculty: Dr. Bernd Chudoba

Department of Mechanical & Aerospace Engineering, University of Texas at Arlington

MAE 4350 – Aerospace Vehicle Design I – Spring 2025

Mission Introduction & Team

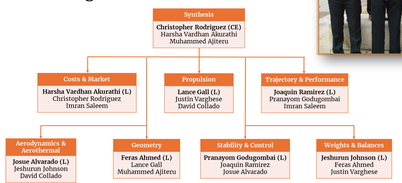
Vehicle: Interplanetary Travel System

- Two Stage to Orbit (TSTO) Launch Vehicle
- Reuseable first stage
- Vertical Takeoff at Boca Chica, TX
- Vertical Landing of the first stage at Boca Chica, TX

Deliverables

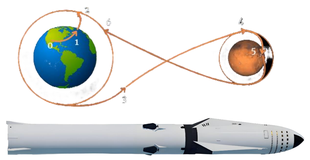
- 3D Solution Space
- 3D Vehicle Model
- Final Team Report

Team Organization Flowchart



Project Initiation

- Identify unknown mission parameters that are crucial for analysis.
- Define key mission parameters and constraints that guide design decisions.
- Generate Mission Trade Matrix serving as a foundational reference for refining the vehicle architecture and optimizing system parameters.



Launch – Low Earth Orbit (LEO)	LEO - Mars	Mars – LEO
Stage Separation Altitude [km]: 0 – 160	Earth – Mars Transfer Method: Hohmann Transfer	Mars – Earth Transfer Method: Hohmann Transfer
LEO Orbital Altitude [km]: 160 – 500	LEO – Mars Distance [km]: 160 – 225E+06	-
Passenger Count: 0 – 100	-	-
Payload [metric tons]: 0 – 200	-	-

Parametric Sizing

Objectives

- Generate & verify solution space of converged vehicles
- Select design point to take into CL

Analysis

- Feed mission requirements & trade variations into AVDS sizing methodology
- Insert disciplinary inputs into hypersonic convergence
- Achieve convergence

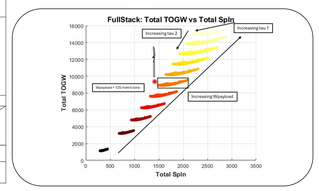
Outputs

- Generate & Verify solution space
- Screen solution space for high-risk / underperforming launch vehicles
- Select design point

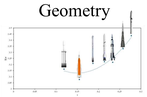
AVDS Sizing (Hypersonic Convergence) is capable of simultaneously sizing launch vehicle components through the iteration of tau (slenderness parameter).

$$W_{OEW} = \frac{W_{str} + C_{sys} + W_{oper} + \frac{(T/W)_{max} W R}{E_{TW}} (W_{pay} + W_{crew})_{max}}{\frac{1}{1+\mu_a} - f_{sys} - \frac{(T/W)_{max} W R}{E_{TW}}}$$

$$W_{OWE} = \frac{\tau \cdot S_{pin}^{1.5}(1 - k_{vp} - k_{vs}) - V_{fix} - V_{pay} - V_{crew}}{\frac{WR}{P_{ppl}} + k_{ve}(T/W)_{max}}$$



Geometry



$K_w = 1.385Sr^3 + 17.291r^2 - 5.286\lambda r + 3.5297$

Propulsion

Parameter	Value	Discipline
$(T/W)_{ij}$ [lf/lf]	1.24	Synthesis
$(T/W)_{ij}$ [lf/lf]	1.43	Synthesis
E_{TW} [lf/lf]	174	Synthesis
K_{vs} [m³/lf]	0.013	Synthesis
ρ_{ppl} [kg/m³]	851	Synthesis

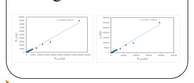
Aerodynamics & Aerothermodynamics

Vehicle	Drag Coefficient	Mach Number	Altitude (km)	Maximum Dynamic Pressure (kPa)
Falcon 9	0.255	1.75	16.100	28.587
New Glenn	0.260	1.525	13.121	25.829
Saturn V	0.500	1.825	13.818	30.000
ITS	0.325	1.825	13.800	29.000

Trajectory & Performance

Vehicle Phase	Parameter	Values
Booster Descent (Mass Accounted during Launch Column)	Weight Ratio 1	12.0816
ITS Starship after separation	Weight Ratio 2	1.9608
ITS Starship during IT	Weight Ratio 3	5.2692
ITS Starship during Mars Descent	Weight Ratio 4	1.6498
ITS Starship during Mars Ascent	Weight Ratio 5	4.8197
ITS Starship during IT	Weight Ratio 6	5.2692
Super Heavy Booster During Descent	Weight Ratio Descent	1.5342

Weights & Balances



Disciplines	Parameters
Propulsion	$\rho_{ppl}, E_{T/W}, k_{ve}, (T/W)_{max}$
Weights & Balances	$C_{sys}, f_{sys}, f_{str}$
Geometry	τ, K_w, S_{pin}
Trajectory	WR
Synthesis	$\mu_a, \rho_{pay}, E_{TW}, k_{vp}, k_{ge}, k_{vs}, k_{pv}$, TOGW, V_{fix}, V_{pay}

Configuration Evaluation

Objectives

- Determine Feasibility & Analyze Fully Configured Vehicle Using Higher Fidelity Methods

Analysis

- Perform Multidisciplinary Flight Phase Analysis
- Modify Vehicle As Required to Meet Disciplinary Requirements

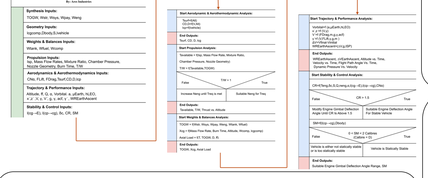
Outputs

- Generate & Verify Redefined Solution Space With Converged Values

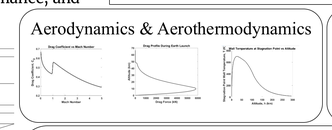
Flight Phase I (Earth Launch – Low Earth Orbit)

Objectives

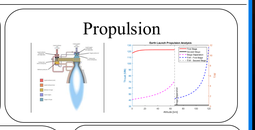
- Ensure Thrust to Weight (T/W), Control Ratio (CR), and Static Margin (SM) Requirements Are Met
- Key Disciplines Include Aerodynamics & Aerothermodynamics, Propulsion, Weights & Balances, Trajectory & Performance, and Stability & Control



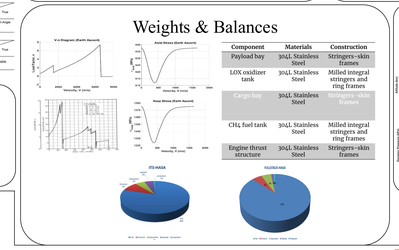
Aerodynamics & Aerothermodynamics



Propulsion

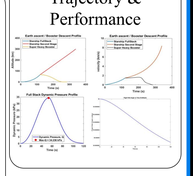


Weights & Balances

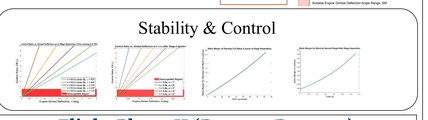


Component	Materials	Construction
Payload bay	304L Stainless Steel	Stringers–skin frames
LOX oxidizer tank	304L Stainless Steel	Milled integral stringers and ring frames
Cargo bay	304L Stainless Steel	Stringers–skin frames
CH4 fuel tank	304L Stainless Steel	Milled integral stringers and ring frames
Engine thrust structure	304L Stainless Steel	Stringers–skin frames

Trajectory & Performance



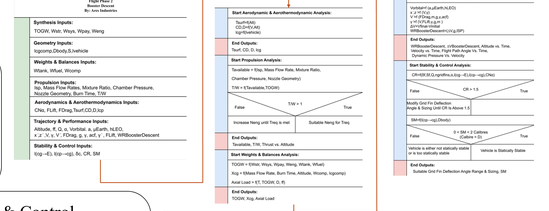
Stability & Control



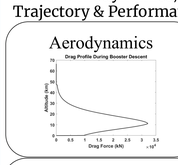
Flight Phase II (Booster Descent)

Objectives

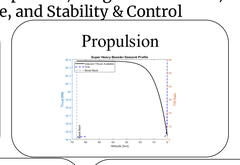
- Ensure Thrust to Weight (T/W), Control Ratio (CR), and Static Margin (SM) Requirements Are Met
- Key Disciplines Include Aerodynamics & Aerothermodynamics, Propulsion, Weights & Balances, Trajectory & Performance, and Stability & Control



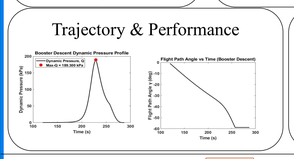
Aerodynamics



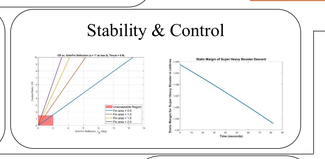
Propulsion



Trajectory & Performance



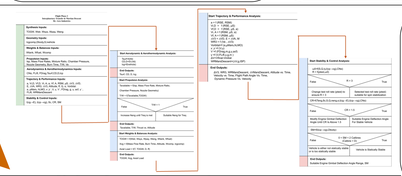
Stability & Control



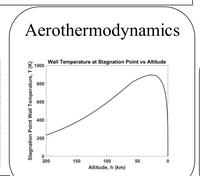
Flight Phase III (LEO – Martian Descent)

Objectives

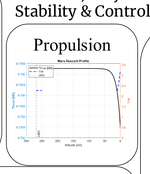
- Ensure Thrust to Weight (T/W), Spin Ratio (R), Control Ratio (CR), and Static Margin (SM) Requirements Are Met
- Key Disciplines Include Aerodynamics & Aerothermodynamics, Propulsion, Weights & Balances, Trajectory & Performance, and Stability & Control



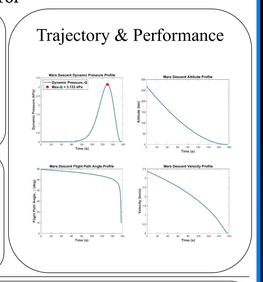
Aerothermodynamics



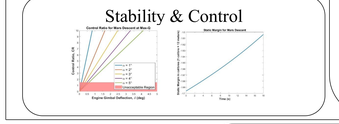
Propulsion



Trajectory & Performance



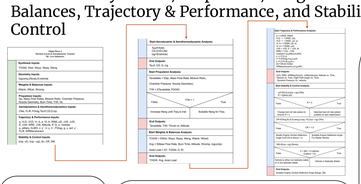
Stability & Control



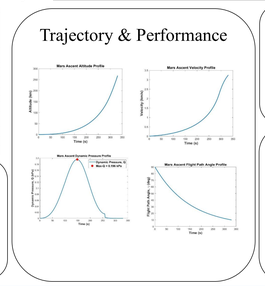
Flight Phase IV (Martian Ascent – LEO)

Objectives

- Ensure Thrust to Weight (T/W), Spin Ratio (R), Control Ratio (CR), and Static Margin (SM) Requirements Are Met
- Key Disciplines Include Aerodynamics & Aerothermodynamics, Propulsion, Weights & Balances, Trajectory & Performance, and Stability & Control

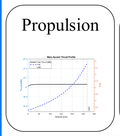


Trajectory & Performance

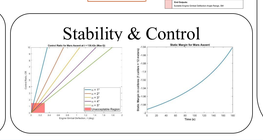


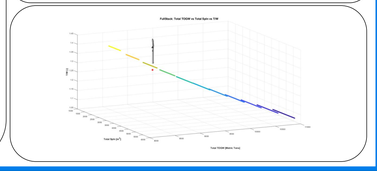
Cost & Market

Propulsion



Stability & Control





Configuration Layout

Synthesis: Configuration Layout
By: Christopher Rodriguez, Harsha Akurathi, Muhammed Ajiteru

Mission Requirements:

- VLVL from Boca Chica, TX
- VLVL on Elysium Planitia, Mars
- 100+ Metric Ton Payload
- Human Life Support

Literature Review:

- Cruyaz, P., A, Bruno, C., and Chudoba, B.,
- Future Spacecraft Propulsion Systems and Integration
- Murthy, S.N.B., and Curran, E.T., Scramjet Propulsion
- Harris C., and Chudoba, B., AVDS Education Lab Module 2
- Harris C., and Chudoba, B., AVDS Education Lab Module 3

Objectives:

- Configure & Justify Internal & External Vehicle Components

Inputs:

- Design Point (Vehicle Weight & Volume) From PS

Multidisciplinary Definitions of Weight/Volume & Locations of Lift, Thrust, Volume & Control Hardware

Stability & Control:
Control Effectors Selection & Placement

Aerodynamic & Aerothermal:
Lifting Surfaces Placement
TPS Material & Placement

Propulsion:
Propulsion System Selection
Engine Number & Flow Paths
Engine Placement

Weights & Balances:
Fuselage Cross Section
Weight Distribution
Internal Hardware Layout & Placement
Landing Gear Selection

Geometry:
Internal Volume Allocation
Internal & External Hardware Modeling

Outputs:

- Finalized & Justified Internal & External Vehicle Configuration
- Generated Model of Finalized Configuration

Objectives

- Configure & Justify Internal & External Vehicle Hardware

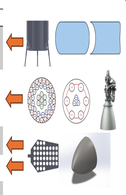
Analysis

- Lifting Surface & TPS Justification & Placement
- Propulsion System Selection & Engine Number & Placement
- Internal Layout & Placement & Landing Gear Selection
- Control Effector Selection & Placement
- Internal & External Hardware Modeling

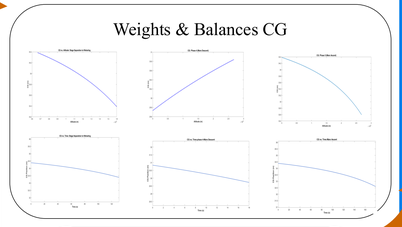
Outputs

- Justified Internal & External Vehicle Configuration
- Generated Vehicle Model of Finalized Configuration for CE Analysis

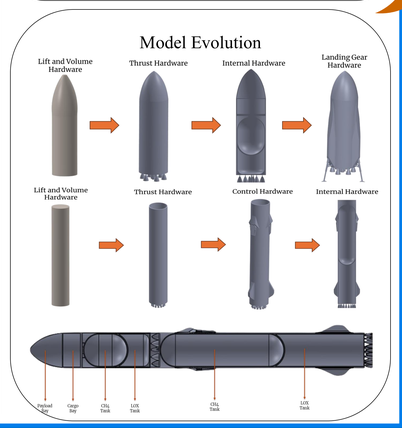
Discipline	Input Needed
Geometry	Landing Gear System Selection Fuel Tank Shape
Propulsion	Engine Number Engine Packing/Spacing Propulsion System Selection
Aerodynamics & Aerothermal	Booster Grid Fin Sizing
Stability & Control	Control Effector Selections
Weights & Balances	Internal Component Placement Payload Location Fuselage Cross Section Wall Thickness



Weights & Balances CG



Model Evolution



References

1. Chudoba, B., and Harris, C., "AVDS Education – Lab Module 3 - PI – PS – CL – CE," University of Texas at Arlington, Jan 29 2025.
2. Industries, A., "Reverse Engineering the SpaceX Martian Starship (Final Presentation)," University of Texas at Arlington, Apr 23 2025.
3. Industries, A., "Ares Industries: Reverse Engineering The SpaceX 2016 Martian Starship)," University of Texas at Arlington, May 8 2025.